

WORKFLOW

Quotes Recommendation Chatbot Using NLP

Overview

The workflow of the “Quotes Recommendation Chatbot Using NLP” project is structured into four major milestones. Each milestone represents a specific phase in the development lifecycle, beginning from problem identification to final deployment and validation. This structured approach ensures systematic development, proper documentation, and efficient implementation of the chatbot system.

MILESTONE 1: DEFINE PROBLEM / PROBLEM UNDERSTANDING

Activity 1: Specify the Business Problem

The primary objective of this project is to design a chatbot that recommends relevant quotes based on user input. Many users seek motivational, inspirational, or theme-based quotes but often struggle to find suitable content quickly. The chatbot aims to solve this problem by understanding user queries through Natural Language Processing (NLP) and recommending appropriate quotes instantly.

Activity 2: Business Requirements

- The chatbot should understand natural language queries.
- It should categorize user input into predefined themes such as motivation, love, success, and life.
- The chatbot must provide accurate and relevant quote recommendations.
- The system should be user-friendly and interactive.
- The solution should be scalable and reusable for future enhancements.

Activity 3: Literature Survey

A literature survey was conducted to study existing chatbot systems and NLP-based recommendation models. Research papers, online resources, and documentation on Rasa and NLP techniques were reviewed. The survey helped in understanding different chatbot architectures, NLP techniques used in intent recognition, and implementation strategies for conversational AI systems.

Activity 4: Social or Business Impact

Social Impact:

- Provides users with quick access to motivational and inspirational content.

- Enhances user engagement and emotional well-being.

Business Impact:

- Can be integrated into websites or applications to increase user interaction.
- Can serve as a content recommendation feature for digital platforms.

MILESTONE 2: ENVIRONMENT SETUP

Activity 1: Install Rasa and Required Dependencies

Rasa framework was selected for chatbot development due to its open-source nature and strong NLP capabilities. Python (version 3.8 or above), Rasa framework, and required dependencies were installed to prepare the development environment.

Activity 2: Setting Up the Rasa Project

After installation, a new Rasa project was initialized. The default project structure includes `nlu.yml` for training data, `domain.yml` for defining intents and responses, `stories.yml` for conversation flows, and `config.yml` for model configuration. This structured setup supports organized chatbot development.

MILESTONE 3: DATA COLLECTION & MODEL BUILDING

Activity 1: Data Collection – Creating User Queries (nlu.yml)

Sample user queries were created under different intent categories such as motivational quotes, love quotes, and success quotes. These examples were added to the `nlu.yml` file to train the model for intent recognition.

Activity 2: Creating Bot Responses (domain.yml)

Predefined responses containing categorized quotes were added in the `domain.yml` file. Each intent was mapped to corresponding responses to ensure accurate output generation.

Activity 3: Defining Interaction Between User and Chatbot (stories.yml)

Conversation flows were defined in the `stories.yml` file. Stories describe how the chatbot should respond when a user provides specific input, ensuring smooth and logical interaction.

Activity 4: Model Training

After preparing the training data and configuration files, the Rasa model was trained. The training process enabled the chatbot to identify user intents, match intents with predefined responses, and provide accurate quote recommendations.

Activity 5: Model Storage and Reusability

Once training was completed, the generated model was stored for future use. Model storage ensures easy deployment, reusability without retraining, and efficient version control.

MILESTONE 4: TESTING & DEPLOYMENT

Activity 1: Model Testing Using Rasa Shell

The chatbot was tested using the Rasa shell to verify intent recognition and response accuracy. This phase helped identify and correct errors in conversation flow.

Activity 2: Testing Using Test Stories

Test stories were created to simulate real-world user interactions, ensuring consistent chatbot performance.

Activity 3: Deployment Using Web Interface

After successful testing, the chatbot was deployed using a web interface, allowing users to interact through a browser-based platform.

Activity 4: Validation of Deployed Chatbot

The deployed chatbot was validated by testing multiple user queries, checking response relevance, and ensuring smooth conversation flow. Successful validation confirmed that the chatbot meets business requirements.

Conclusion

The workflow of the “Quotes Recommendation Chatbot Using NLP” project follows a structured milestone-based approach. Starting from problem definition to final deployment, each phase contributes to building an efficient and functional chatbot system. This systematic workflow ensures clarity, maintainability, and scalability while aligning with SmartBridge internship standards.